Compact-2D: A Physical Design Methodology to Build Commercial-Quality Face-to-Face-Bonded 3D ICs

Bon Woong Ku, Kyungwook Chang, and Sung Kyu Lim
School of ECE, Georgia Institute of Technology, Atlanta, GA
{bwku,k.chang,limsk}@ece.gatech.edu

ABSTRACT

The recent advancement of wafer bonding technology offers fine-grained and silicon-space overhead-free 3D interconnections in face-to-face (F2F) bonded 3D ICs. In this paper, we propose a full-chip RTL-to-GDSII physical design solution to build high-density and commercial-quality two-tier F2F-bonded 3D ICs. The state-of-the-art flow named Shrunk-2D (S2D) [10] requires shrinking of standard cells and interconnects by a factor of 50% to fit into the target 3D footprint of a two-tier design. This, unfortunately, necessitates commercial place/route engines that handle one node smaller geometries, which can be challenging and costly. Our flow named Compact-2D (C2D) does not require any geometry shrinking. Instead, C2D implements a 2D IC with scaled interconnect RC parasitics, and contracts the layout to the F2F design footprint. In addition, C2D offers post-tier-partitioning optimization that is shown to be effective in fixing timing violations caused by inter-tier 3D routing, which is completely missing in S2D. Lastly, we present a methodology to recycle the routing result of post-tier-partitioning optimization for final GDSII generation. Our experimental results show that at iso-performance, C2D offers up to 26.8% power reduction and 15.6% silicon area savings over commercial 2D ICs without any routing resource overhead.

CCS CONCEPTS

• **Hardware → 3D integrated circuits**; **Physical design (EDA)**;

KEYWORDS

Compact-2D; Physical Design Methodology; Wafer-level Bonding; Face-to-Face (F2F) Bonded 3D ICs

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1 INTRODUCTION

Face-to-face (F2F) bonding technology involves 3D integration of two pre-fabricated dies in a face-to-face fashion. In F2F bonding, electrical connections between the dies are made by F2F vias, and the minimum pitch of these F2F vias defines the density of 3D interconnections. As 2D interconnects become denser along with logic device scaling, it calls for a tighter 3D interconnect pitch to improve the functional density and power-performance-area benefit of F2F-bonded 3D ICs. To enable smaller F2F via pitches, R&D has focused on enhancing the bonding precision of F2F integration lately. Among several notable achievements, hybrid wafer-to-wafer (W2W) bonding technology has emerged as a promising solution.

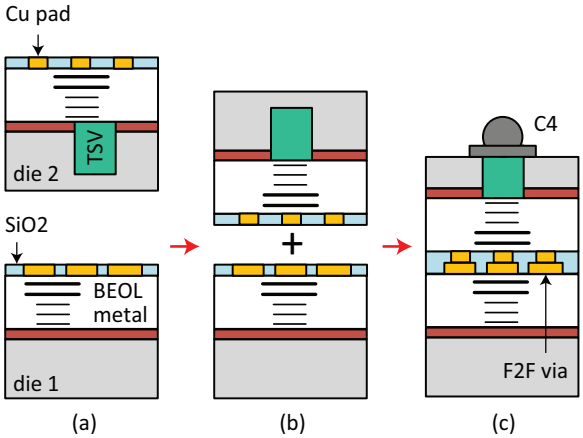

Figure 1: F2F integration using hybrid W2W bonding technology [3]. (a) Wafers are fabricated in parallel before bonding. One wafer contains via-middle TSVs for the I/Os. The wafer surface is flattened by chemical-mechanical planarization. (b) Wafers are aligned, bonded at room temperature, and annealed at < 250 °C. (c) F2F vias are formed at the locations of direct metal-to-metal bonding. The top wafer is thinned to reveal TSVs for bumping.

Hybrid W2W bonding is a wafer-level integration technology that enables direct metal-to-metal (damascene-pad) and dielectric-to-dielectric bonding between the back-end-of-lines (BEOLs) of pre-fabricated wafers [9, 12]. After an annealing process at a low temperature (< 250 °C) to strengthen the inter-facial bonding, F2F vias are naturally formed at the locations of direct metal-to-metal bonding. Thanks to the high precision of wafer-level integration, a $1.8\mu m$ F2F via pitch has been demonstrated. In addition, the minimum pitch is projected down to $0.8\mu m$ in the near future [3, 6]. This allows designers to utilize fine-grained and silicon-space overhead-free 3D interconnections in F2F-bonded 3D ICs.

Various studies have shown the benefits of F2F-bonded 3D ICs. Using a $5\mu m$ F2F pitch, [5] demonstrated a test chip that achieves a high memory bandwidth ($63.8GB/s$) in core-memory stacking

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architecture at $4W$ power consumption. [11] adopted F2F bonding technology for the heterogeneous integration of MEMS and SoCs, and reported 30% form factor savings. However, all these benefits are still based on a large F2F via pitch. To fully benefit from the advanced F2F integration technology, new design and CAD solutions are required. In this paper, we present a physical design methodology named Compact-2D (C2D) to build high-density and commercial-quality F2F-bonded 3D ICs.

2 MOTIVATION

The state-of-the-art in F2F full-chip design flow, Shrunk-2D (S2D) [10], proposed a unique method to build F2F-bonded 3D ICs. S2D requires shrinking of standard cells and interconnects by 50% to fit into the footprint of a two-tier F2F design with no silicon-area overhead. Then, the shrunk layout objects are used to implement the Shrunk-2D design, where the (X,Y) locations of cells are optimized with the same half perimeter wirelength (HPWL) as that of the target F2F design, assuming that the Z dimension is so small and thus negligible. To decide the Z location of each cell, tier partitioning is subsequently performed. Then, F2F via planning decides the actual F2F via locations based on the (X,Y,Z) placement solution. Although S2D shows how to use commercial 2D P&R engines to design F2F-bonded 3C ICs, it introduces the following new issues, especially in the advanced technology nodes.

- To handle shrunk geometries, S2D requires place/route (P&R) engines and design rule checkers that target one node smaller technology, which is both challenging and costly.[1]
- The shrunk dimension of interconnects leads to inaccuracy in RC parasitics of the S2D design unless the parasitic database is rebuilt for the shrunk geometries.[2]
- Tier partitioning in S2D ignores the fact that any inter-tier 3D route requires the *full* metal stacks for *both* tiers in F2F designs. Nevertheless, S2D does not support any optimization after tier partitioning. Therefore, it is prone to timing failure caused by inter-tier 3D routing overheads.

The physical design of monolithic 3D ICs [4, 7, 8] resembles that of F2F-bonded 3D ICs because the inter-tier vias are negligibly small. This offers similar freedom in constructing a (X,Y,Z) placement solution for both monolithic 3D and F2F designs. However, a notable difference lies in how inter-tier routing is done: in monolithic 3D ICs, only a single stack of BEOL is used, whereas both stacks are required in F2F-bonded 3D ICs. This motivates us to address the inter-tier 3D routing overheads efficiently (in both timing and power) for commercial-quality F2F designs. Thus, existing works on monolithic 3D ICs cannot be easily migrated to handle F2F designs.

3 DESIGN METHODOLOGY

This section presents our design methodology named Compact-2D (C2D) flow to build commercial-quality F2F-bonded 3D ICs. C2D flow finds the (X,Y) placement solution of a F2F design using the original geometries of standard cells and interconnects. It also introduces an optimization capability to take the inter-tier 3D routing overheads into account correctly. The overall design methodology is shown in Figure 2.

[1]Our conversations with S2D flow users at industry design houses revealed an exponential increase in design rule violations at the 7nm node. S2D suggests that designers ignore these errors. However, they reported that an excessive number of violations may cause commercial engines to terminate abruptly or produce low-quality layouts.
[2]Unless the resistivity and thickness of an interconnect are modified, the unit length resistance of a wire segment is not the same in S2D and F2F designs because the width of the interconnect is shrunk by 29.3%. Similarly, the shrunk width and spacing of the interconnect lead to inaccurate capacitance values in S2D designs.

Table 1: Terminologies in our Compact-2D (C2D) flow.

Compact-2D Design	An initial 2D design with unit length RC scaled by a factor of 0.707
Memory Expansion	Expanding the pin locations and memory macro boundaries by a facror of 1.414
Placement Contraction	Linearly contracting the placement solution of a Compact-2D design by a factor of 0.707
Compact F2F Via Planning	Performing timing, power, and F2F via location co-optimization to address inter-tier routing overhead
Incremental Routing	Recycling the routing result from Compact F2F Via Planning for the final GDSII generation

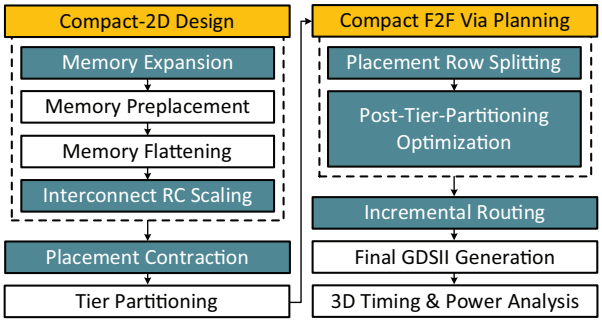

Figure 2: Our Compact-2D (C2D) flow. In color are the key steps proposed in this paper to build commercial-quality F2F-bonded 3D ICs using 2D IC implementation tools.

3.1 Compact-2D Design

A Compact-2D design is a pseudo-3D design in C2D flow to find the optimal (X,Y) locations of standard cells in a target F2F design. The floorplan of the Compact-2D design is two times as large as the final 3D footprint in the same aspect ratio to accommodate all the synthesized gates in the two-tier F2F design with their original geometries. However, the HPWL of a net in the F2F design is 29.3% shorter than the corresponding net in the Compact-2D design when both are projected on the X-Y plane. To match the electrical length despite the difference in geometrical length, Figure 3 illustrates the need for interconnect RC scaling in the Compact-2D design. By Scaling the unit RC per length by a factor of 0.707, we avoid the redundant buffer insertions caused by increased geometrical length in the Compact-2D design while still using the original geometries of standard cells and interconnects. Then, we perform all the required implementation steps of the conventional 2D ICs in the Compact-2D design.

3.2 Placement Contraction

Once the Compact-2D design is implemented, the cell locations are linearly mapped to the 3D design footprint to finalize the optimal

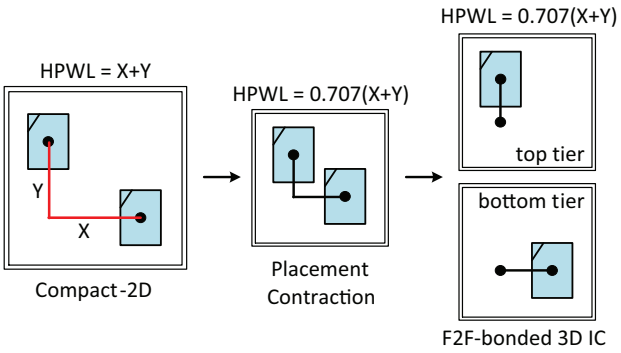

Figure 3: The need for interconnect RC scaling in a Compact-2D design. The length of interconnects will be reduced to 0.707X in the final F2F layout. In order to reflect this, we reduce the unit length RC to 0.707X in the Compact-2D design. The red line in the most left figure indicates an interconnect with reduced parasitics.

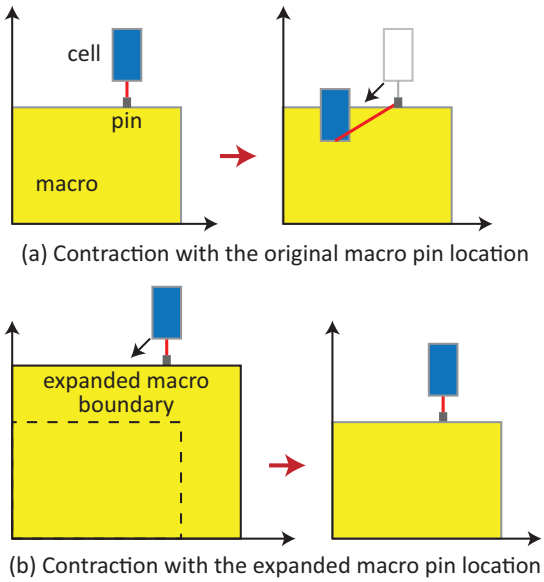

Figure 4: The need for the expansion of memory boundaries in C2D flow. (a) The original macro pin location causes placement contraction to introduce unwanted routing change and cell overlap, (b) The macro boundary and its pin locations are expanded by a factor of 1.414 to resolve this issue.

(X,Y) locations of cells in the F2F design. This is called placement contraction. Considering the linearly contracted HPWL of a net, the scaled interconnect RC parasitics of the Compact-2D design are the same as those of the F2F design in the original unit length RC. Also, it implies that the (X,Y) solutions based on the shrinking idea from S2D and interconnect RC scaling / placement contraction ideas from our C2D are ideally the same. However, C2D necessitates the P&R engines that handle the target technology node only while S2D relies on the CAD engines for the next technology node.

3.3 Handling Memory Macros

In the conventional 2D IC design, memory macros are preplaced in the floorplan without any overlaps, and none of standard cells is placed inside the memory macros. However, the Compact-2D design needs to allow overlaps of memory macros when they share the same (X,Y) location, but at different Z locations. Moreover, P&R engines should be allowed to place the standard cells inside the memory macro regions unless the memory macros fully occupy the regions in both tiers. Previously, S2D proposed shrinking the footprint of memory macros down to the minimum placement unit, and use placement blockages at its original boundary. Full placement blockages are used in the fully overlapped regions of memory macros, and restrict the standard cell placement. To enable the standard cell placement at partially vacated regions, 50% partial placement blockages are used. The pin locations of memory macros are retained, which serve as anchors for the standard cell placement regardless the footprint change of memory macros.

C2D follows the same way, but requires an additional step. Considering that the boundary of placement blockages should be the same as the original boundary of memory macros after placement contraction, the placement blockages for memory macros needs to be expanded by a factor of 1.414 for the Compact-2D design. The pin locations of memory macros also should be expanded to correctly anchor the standard cells around the placement blockages as shown in Figure 4. Therefore, at the floorplan stage of the Compact-2D design, we should prepare for the expanded memory macro Library Exchange Format (LEF) files (Memory Expansion), assign their tier locations, and preplace them manually considering the inter-module connectivity (Memory Preplacement), and generate placement blockages on the expanded memory regions while flattening the tier locations of memory macros (Memory Flattening).

3.4 Tier Partitioning

Since the tier locations of memory macros are preassigned manually, the standard cells within the memory macro boundaries move to the tier where memory macros do not occupy. To determine the Z location of each standard cell outside the memory macro boundaries, C2D introduces tier partitioning that utilizes bin-based placement-driven Fiduccia-Mettheyses (FM)-mincut partitioning algorithm [10]. Each partitioning bin is defined in a regular fashion on the final F2F footprint, and we run the algorithm based on the (X,Y) solution derived from placement contraction.

Bin-based placement-driven FM-mincut partitioning helps balance the area skew over the entire design footprint, otherwise resulting in huge white spaces or displacement from the optimal (X,Y) locations during placement legalization. The number of cutsize, which turns into the minimum number of inter-tier connections, is controlled by the size of partitioning bins. Too many cutsize leads to routing congestion, while too few cutsize decreases the power-performance benefits of F2F-bonded 3D ICs. Therefore, a sweet spot exists along the partitioning-bin size. Once tier partitioning determines the Z location of each cell, a placement engine legalizes the overlaps caused by placement contraction, and a Design Exchange Format (DEF) file for each die is created.

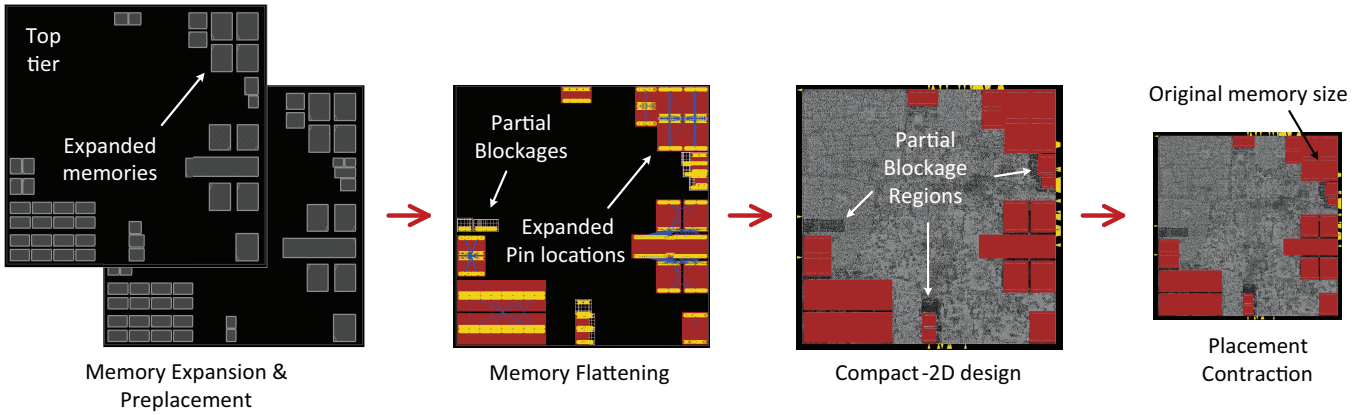

Figure 5: Our C2D flow demonstrated with OpenSparc T2 [1] single core design: memory expansion and preplacement, memory flattening, Compact-2D design, and placement contraction. Tier partitioning and Compact F2F via planning follow next.

3.5 Compact F2F Via Planning

After we decide the (X,Y,Z) locations of standard cells, we should determine the F2F via locations. This is called F2F via planning. In this step, inter-tier 3D routing overhead, which is not accounted by the Compact-2D design, starts to affect the design closure. S2D is not only susceptible to this degradation, but none of 3D-routing-aware optimization is introduced after tier partitioning. In order to support post-tier-partitioning optimization (post-TP opt) to compensate the inter-tier routing overhead, C2D presents a unique stage named Compact F2F via planning. Compact F2F via planning consists of two steps, and following subsections describe them in detail.

3.5.1 Placement Row Splitting. Compact F2F via planning performs based on the 3D technology LEF which includes the definition and design rules of metal stacks in both tiers. 3D macro LEFs are required for the commercial router to distinguish the pin layer of macros based on their tier locations. Next, our in-house program creates a DEF and a Verilog file by instantiating the cells with 3D macro LEFs while flattening the tier location of cells. However, in order to fully utilize the optimization capabilities, the flattened DEF file should not have the placement overlaps although all synthesized gates are accommodated in the final F2F design footprint. Therefore, we split a placement row into the top and bottom rows, and change the height of standard cells in 3D macro LEFs to the half of the original to fit into the split rows.

In Figure 6, Row0 and Row1 are two adjacent placement rows to be split. Row1 is vertically flipped over to share the power rail with Row0. Now, placement row splitting turns each row into two horizontally split rows. In Row0, the bottom half is reserved for the bottom tier placement, and the top half for the top tier. However, in Row1, the bottom half is reserved for the top tier placement and the top half for the bottom tier due to the flipped orientation of Row1. As a result, the placement overlap is fully legalized while accommodating every cell in the design on the final F2F footprint. It is worth noting that the pin locations of standard cells are preserved regardless of splitting placement rows. Based on the retained pin locations and the same width of standard cells, accurate post-TP opt proceeds.

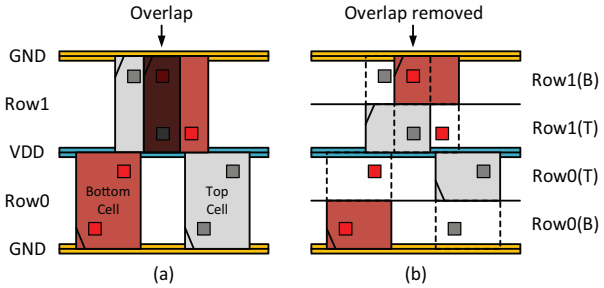

Figure 6: (a) Shrunk-2D flow [10] does not offer post-tier-partitioning optimization because of the placement overlap. (b) Placement row splitting in our C2D flow enables the optimization by fully legalizing the placement overlap.

3.5.2 Post-Tier-Partitioning Optimization. C2D performs timing, power, and F2F via location co-optimization to close the design under inter-tier 3D routing overhead. Post-TP opt requires RC corners for the full 3D metal stack including the F2F via, and the timing corners for both top and bottom tiers. Thanks to placement row splitting, full optimization capabilities, including insertion, deletion, move, and resizing, are employed. Once the optimization is done, our in-house binaries create a DEF file for each die that introduces the F2F vias as I/O ports (F2F ports) and contains the final cell locations by restoring the original cell height. Since the pin locations of cells are preserved regardless of having the cell height, we can easily retrieve the correct (X,Y) locations of cells based on the original cell height. Also, we generate a Verilog file for each die that presents the connectivity among F2F ports and the cells within a die. A top Verilog file that defines the connections between F2F ports in separate tiers is created, and lastly, we generate a top Standard Parasitic Exchange Format (SPEF) file that presents the RC parasitics of F2F vias.

3.6 Incremental Routing

Incremental routing is a CAD solution to preserve the routing result of Compact F2F via planning for the final GDSII file generation of

each die. We first construct a graph for each net that consists of vertices and edges representing individual routing objects and their connectivities. Routing objects include a wire, a via, an I/O port, or a cell pin. The X/Y locations where those routing objects cross each other are kept along with their edge definitions. Next, if the graph contains a F2F via vertex, we convert it into two vertices without an edge between them representing I/O ports for the top and bottom tier. Each vertex is only connected to the adjacent vertices that shares the same tier. As a result, the graph turns into a group of disconnected subgraphs, and each subgraph represents a 2D subnet on the specific die. Now, we reproduce the routing result for each subgraph based on the actual connection points defined in each edge. A depth-first search ordering is used to make the output in the format of DEF syntax. Finally, the routing information for each subgraph is delivered to the DEF for a corresponding die.

In the final GDSII file generation step, we use this routing information as an initial solution for sign-off physical design rule violation (DRV) fixing. The reason why DRV fixing is necessary is that tools built for 2D ICs do not support full DRV fixing for the pins outside the macro boundary while employing placement row splitting. When the sign-off DRV fixing is done, RC parasitics of each die are extracted, and we proceed the final 3D timing & power analysis.

4 STATE-OF-THE-ART COMPARISON

In Table 2, we compare the timing & power savings of C2D with those of S2D based on the OpenSparc T2 [1] single core (SPC) design at $1.0GHz$ clock frequency. We use dual-Vt cell libraries in 28nm commercial-grade technology process design kit (28nm PDK). Six metal layers are used for 2D, and the top and bottom tiers for F2F implementations. The F2F via diameter, pitch, resistance and capacitance are assumed to be $0.5\mu m$, $1.0\mu m$, 0.5Ω, and $0.2fF$, respectively. For the static power analysis, we set the switching activity as 0.1 for primary input ports and register output pins, and 2.0 for a clock port.

We observe that both C2D and S2D designs significantly decrease the net switching power thanks to the huge wirelength savings in F2F designs. Following buffer reduction contributes to the cell internal power savings. The total power reduction of C2D is 11.3% while S2D offers a 11.0% savings over 2D IC at iso-performance. In addition, it is remarkable that C2D reduces the total negative slack violations by 83.6% while S2D worsens the timing. This result not only shows that C2D offers comparable power reduction as the state-of-the-art S2D, but also proves that C2D builds timing-robust F2F designs. Most of all, C2D is more scalable than S2D in that our C2D flow performs with P&R engines, technology files, and design rules for the target technology and does not require handling of the next smaller node.

5 EXPERIMENTAL RESULTS

In this section, we analyze the impact of each design step in C2D flow with LDPC, AES-128, and JPEG from OpenCore benchmark suites [2]. Assumptions on the technology and analysis are the same as Section 4 made. The initial utilization density for AES-128 and JPEG is 60%, while 40% for wire-dominated LDPC. The maximum clock frequency for each benchmark is 2.0GHz for LDPC, 5.4GHz for AES-128, and 2.16GHz for JPEG. Figure 7 shows the GDSII layouts of 28nm 2D and C2D-based F2F implementations for each benchmark including SPC at their maximum frequency.

Table 2: Timing & power comparison among 2D, S2D [10], and C2D using OpenSparc T2 [1] single core (28nm). Δ% shows % improvement over 2D. Target clock period is 1ns. C2D offers comparable power reduction and significant performance savings compared with S2D.

	2D	S2D	Δ%	C2D	Δ%
Total WL (m)	15.36	11.77	23.4%	11.55	24.8%
F2F Via #	-	154,127	-	193,487	-
Footprint (mm^2)	2.53	1.26	50.2%	1.26	50.2%
Total Power (mW)	338.20	300.87	**11.0%**	299.88	**11.3%**
Cell Power (mW)	82.12	79.11	3.7%	79.07	3.7%
Net Power (mW)	183.26	153.33	16.3%	150.86	17.7%
Leak. Power (mW)	72.83	68.43	6.0%	69.95	4.0%
Mem. Power (mW)	45.98	44.94	2.3%	44.77	2.6%
Comb. Power (mW)	171.30	140.90	17.7%	139.90	18.3%
Reg. Power (mW)	67.72	67.68	0.1%	69.80	-3.1%
Clk Tree Power (mW)	53.17	47.34	11.0%	45.40	14.6%
Worst Neg. Slack (ps)	-27.65	-52.52	**-89.9%**	-25.99	**6.0%**
Total Neg. Slack (ps)	-832.85	-846.94	**-1.7%**	-136.75	**83.6%**
Total Pos. Slack (ps)	35988.60	38884.50	8.0%	39422.20	9.5%

5.1 Impact of Interconnect RC Scaling

In a Compact-2D design, we scale interconnect RC parasitics by a factor of 0.707 to imitate the parasitics of wirelength in the final F2F design based on that the footprint of the F2F design is exactly 50% of the 2D footprint. However, the RC scaling factor can be generalized and set to be 0.6 in case the F2F design footprint is 36% of the 2D footprint. Table 3 shows Compact-2D design results with various 3D/2D footprint ratios.

With a low RC scaling factor, such as 0.548, all benchmarks has huge power and standard cell area savings because of the reduced interconnect parasitics and the less number and lower drive-strength of buffers. However, since the target footprint is way smaller than the standard cell area savings, it results in the impractical placement utilization per each die in the F2F design. Assuming placement utilization in [70%, 80%] range is allowed, our footprint savings reach up to 65% for LDPC, and 56% for both AES-128 and JPEG. In case of wire-dominated LDPC design, since the 2D footprint is determined by the routability, the huge wirelength reduction in the F2F design helps increase the footprint savings more.

When the same placement utilization in both 2D and F2F-bonded 3D ICs should be considered, we observe that 53-57% footprint savings are good target for all designs due to the buffer savings from the interconnect RC scaling. With a constraint on the exact 50% footprint savings, we find that 4-12% of placement utilization savings in F2F designs. In summary, sweeping the interconnect RC scaling helps to set the practical and optimal F2F design assumption. This also shows that C2D is incredibly flexible to design and find the optimal footprint of F2F designs for logic benchmarks thanks to the usage of original geometries for standard cells. For the rest of experiments, we keep the 50% footprint savings in F2F designs for all benchmarks to factorize the impact of other steps clearly.

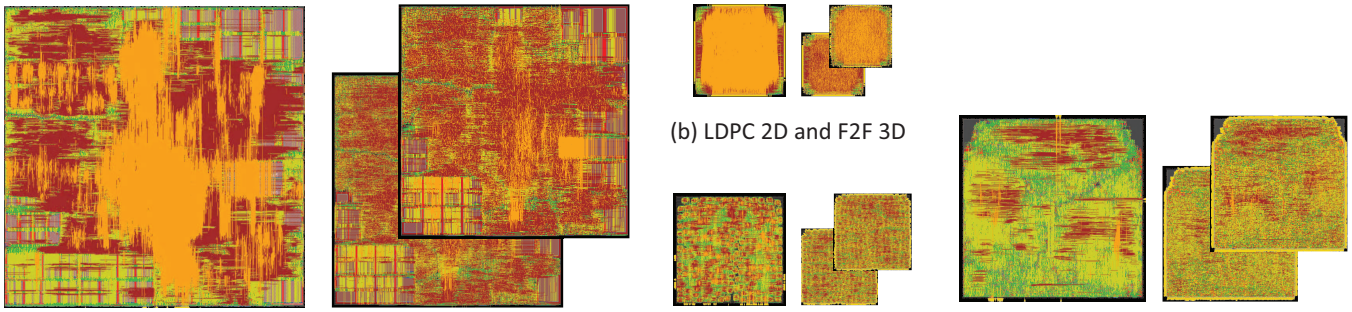

Figure 7: 28nm GDSII die images of 2D and F2F-bonded 3D implementations using our C2D flow. (a) SPC (1.0GHz), (b) LDPC (2.0GHz), (c) AES-128 (5.4GHz), (d) JPEG (2.16GHz).

Table 3: Impact of target 3D footprint. Assuming placement utilization in [70%, 80%] range is allowed, our footprint savings reach up to 65% for LDPC, and 56% for both AES-128 and JPEG

Footprint ($3D/2D$)	50%	45%	40%	35%	30%
RC Scaling	0.707	0.671	0.632	0.592	0.548
LDPC					
Std. Cell Area (mm^2)	0.180	0.178	0.177	0.172	0.169
3D Place. Util. per Die	58.31%	63.92%	72.03%	**79.69%**	91.29%
Place. Util ($3D/2D$)	87.83%	96.30%	108.50%	120.04%	137.51%
Total Power (mW)	179.23	174.48	167.70	158.03	153.85
Footprint ($3D/2D$)	50%	47%	44%	41%	38%
RC Scaling	0.707	0.686	0.663	0.640	0.616
AES-128					
Std. Cell Area (mm^2)	0.359	0.356	0.355	0.355	0.355
3D Place. Util. per Die	70.10%	73.88%	**78.99%**	84.58%	91.43%
Place. Util ($3D/2D$)	95.09%	100.22%	107.15%	116.15%	124.03%
Total Power (mW)	331.68	330.49	324.54	323.39	322.18
JPEG					
Std. Cell Area (mm^2)	0.943	0.941	0.939	0.936	0.933
3D Place. Util. per Die	70.71%	70.71%	**80.07%**	85.65%	92.16%
Place. Util ($3D/2D$)	96.03%	101.78%	108.73%	116.32%	125.15%
Total Power (mW)	579.17	573.52	565.84	563.80	560.10

5.2 Impact of Tier Partitioning

While placement contraction is deterministic in that the (X,Y) locations of cells are scaled by 0.707, bin-based tier partitioning is heuristic w.r.t the size of partitioning bins. Depending on the partitioning-bin size, the number of cells applied to the algorithm varies, resulting in different cut sizes between the dies. Table 4 shows how the different partitioning-bin sizes change the number of 3D connections (F2F vias) and the wirelength of a design. F2F utilization indicates the F2F via usage out of the maximum available number of F2F vias inside the F2F design footprint. While the small bin size leads to the large number of F2F vias, the large bin size allows the algorithm to find the minimum cut size.

To explore the impact of the different number of 3D connections on the wirelength savings, a net is defined as either a 2D or a 3D net based on their F2F usage, and compare its wirelength with that in the Compact-2D design. We observe that the average wirelength per net is correlated to the optimal partitioning-bin size. If the bin size is way smaller ($5\mu m$) than the average net wirelength, most of the nets become 3D, causing congestion and detour to meet the design rules for F2F vias. This is the reason that the wirelength savings of 3D nets decreases at $5\mu m$ bin, lowering the total wirelength savings. On the other hand, if the bin size is too large, then most of the nets remain at 2D, requiring huge legalization caused by placement contraction. Therefore, embracing too much 2D nets again degrades the wirelength savings. LDPC shows the best wirelength savings (27.3%), which is almost ideal (29.3%), when the bin size is in the range of 20 to $80\mu m$, while AES-128, and JPEG, which has short wirelength per net (gate-dominant), have 22.15% and 20.47%, respectively at $10\mu m$ bin. It is noteworthy that gate-dominant circuits steeply loose the wirelength savings along with increasing the bin size over the sweet spots. We determine the size of partitioning bins as $40\mu m$ for LDPC, $10\mu m$ for both AES-128 and JPEG, and proceed Compact F2F via planning.

5.3 Impact of Compact F2F Via Planning

Using LDPC, Table 5 demonstrates that how negatively inter-tier 3D routing affects the timing, and how effectively post-TP opt fixes the timing violations. Since the Compact-2D design does not account the inter-tier routing overheads when it is implemented, we observe that the worst negative slack (WNS) is degraded to 5.87x, and 7.71x for the total negative slack (TNS) after the inter-tier 3D routing is done. All of these timing violations are fixed after we perform post-TP opt. The WNS is improved by 44.4% and the TNS is restored by 91.5% with the negligible power overhead. This proves that post-TP opt in Compact F2F via planning is critical to implement timing-robust F2F designs. In general, post-TP opt restores the timing by inserting or up-sizing the buffers while minimizing the power increase. However, if the power overhead becomes the issue, then post-TP opt can start to delete or down-size the buffers at the expense of the timing margin.

5.4 Impact of Incremental Routing

Table 6 shows how final DRV fixing and tier-by-tier 2D routing affects the design result from post-TP opt, and how much better our incremental routing performs than the existing iterative tier-by-tier

Table 4: Impact of tier partitioning bin size. Smaller bins cause more F2F vias to be used and tend to improve WL saving for 3D. Saving values are w.r.t. 2D results.

Bin Size (μm)	5	10	20	40	80
LDPC					
Bin #	6,169	1,542	386	96	24
Avg. Cell # / Bin	11	42	169	677	2,707
F2F Via #	55,468	26,999	20,850	19,802	19,726
F2F Util. (%)	34.20	16.65	12.86	12.21	12.16
Avg. WL / net (μm)	39.16	38.85	38.83	38.84	38.82
3D Net # (%)	61.41	24.71	17.73	16.89	16.75
3D Net WL Savings (%)	26.73	27.58	27.87	27.87	27.95
2D Net WL Savings (%)	26.60	26.93	26.80	26.79	26.77
Total WL Savings (%)	26.70	27.28	27.32	27.30	27.33
AES-128					
Bin #	10,247	2,562	640	160	40
Avg. Cell # / Bin	14	55	219	877	3,507
F2F Via #	104,306	61,902	51,460	22,311	10,824
F2F Util. (%)	39.16	23.24	19.32	8.38	4.06
Avg. WL / net (μm)	16.45	16.24	16.56	18.16	18.83
3D Net # (%)	59.67	28.11	22.91	11.14	5.96
3D Net WL Savings (%)	20.57	22.10	21.50	18.45	16.73
2D Net WL Savings (%)	22.74	22.20	19.95	11.46	8.76
Total WL Savings (%)	21.14	22.15	20.60	12.94	9.71
JPEG					
Bin #	26,680	6,670	1,668	417	104
Avg. Cell # / Bin	11	43	171	682	2,729
F2F Via #	240,301	120,921	94,868	71,353	53,810
F2F Util. (%)	35.17	17.70	13.88	10.44	7.88
Avg. WL / net (μm)	14.54	14.57	14.76	15.06	15.67
3D Net # (%)	61.36	25.19	18.42	13.27	10.10
3D Net WL Savings (%)	20.69	21.73	21.61	21.39	19.11
2D Net WL Savings (%)	19.76	19.01	17.66	15.53	12.17
Total WL Savings (%)	20.47	20.31	19.29	17.60	14.28

Table 5: Impact of post-tier-partitioning optimization. Δ% indicates its savings. Inter-tier 3D routing (A vs. B) introduces huge timing violations, and our optimization (B vs. C) fixes the timing violations with the negligible power overhead.

Design	LDPC			
Stage	Before 3D Routing (A)	After 3D Routing NO-Opt (B)	After 3D Routing YES-Opt (C)	Δ%
Total Cell (#)	65,187	65,187	65,271	-0.1
Worst Neg. Slack (ps)	-7.42	-43.57	-24.23	44.4
Total Neg. Slack (ps)	-341.86	-2637.13	-222.99	91.5
Total Pos. Slack (ps)	19194.40	17042.80	27072.40	58.8
Violated Path (#)	20	383	27	93.0
Total Power (mW)	179.23	178.25	178.49	-0.1

routing method in S2D. Iterative routing starts the tier-by-tier routing from scratch on top of the placement result of post-TP opt. This leads to a different routing result from post-PT opt due to the final DRV fixing, and perturbs the design closure. We observe that the worst negative slack is degraded to 1.86x, and 25.88x for the total negative slack after using iterative routing. However, our incremental routing preserves the worst negative slack in the acceptable level (less than 25ps under 0.5ns clock period), and retains the total wirelength and power results close to the optimization result (less than 1% overheads).

Table 6: The impact of Final DRV fixing and tier-by-tier 2D routing after post-TP opt. We note that the incremental routing (Incr-R) used in C2D preserves the timing closed by post-TP opt (A vs. C) better than the iterative routing (Iter-R) in S2D [10] (A vs. B). Incr-R also offers smaller wirelength and power overheads for the tier-by-tier routing than Iter-R. Δ% indicates the savings from Incr-R over Iter-R.

Design	LDPC			
Stage	Before 2D Routing (A)	After 2D Routing Iter-R (B)	After 2D Routing Incr-R (C)	Δ%
Total WL (m)	2.721	2.754	2.750	0.1
Worst Neg. Slack (ps)	-24.23	-45.17	-25.16	44.3
Total Neg. Slack (ps)	-222.99	-5771.74	-1599.73	72.3
Total Pos. Slack (ps)	27072.40	11257.00	15107.10	34.2
Violated Path (#)	27	734	402	45.2
Total Power (mW)	178.49	179.53	179.15	0.2

Table 7: Runtime comparison (in minutes): Intel(R) Xeon(R) CPU E5-2640 @ 2.50GHz, 16 cores usage for Cadence Innovus run.

Design	LDPC			AES-128			JPEG		
Runtime (min)	2D	S2D	C2D	2D	S2D	C2D	2D	S2D	C2D
Placement	3	3	3	3	3	3	7	7	7
Pre-CTS Opt.	44	19	22	33	29	28	59	54	55
Clock Tree Syn.	3	5	3	5	6	5	15	17	13
Post-CTS Opt.	8	6	6	12	9	7	15	12	12
Routing	6	8	6	5	7	5	9	11	8
Post-route Opt.	11	10	10	8	8	8	20	19	19
Place. Contr.	-	-	1	-	-	1	-	-	2
Tier Part.	-	1	1	-	3	3	-	11	11
F2F Via Plan.	-	10	10	-	10	10	-	19	19
Post-TP Opt.	-	-	20	-	-	15	-	-	39
Iter. Routing	-	11	-	-	12	-	-	20	-
Incr. Routing	-	-	7	-	-	7	-	-	11
Signoff Analysis	2			3			10		
Final Total	77	75	91	69	90	95	135	180	206

5.5 Runtime Analysis

In Table 7, we tabulate runtime for each design step of 2D, S2D, C2D flows to build LDPC, AES-128, and JPEG. Intel(R) Xeon(R) CPU E5-2640 @ 2.50GHz is used, and 16 cores are employed while running Cadence Innovus. Thanks to the reduced interconnect loads and HPWL savings, Compact-2D designs take 34% less time than 2D until the post-route optimization is done (Compact-2D designs take 10% less time than Shrunk-2D designs at best). However, the total runtime of C2D is longer than that of 2D by a maximum 50% (JPEG), due to the additional steps starting from placement contraction. Although incremental routing achieves a huge runtime savings up to 60% compared with iterative routing in S2D, post-TP opt takes a large portion of the F2F design flow, resulting in 21% runtime overhead in C2D over S2D at worst.

5.6 Commercial 2D vs. C2D

Based on the optimal footprint derived from Section 5.1, we compare the design results of commercial 2D with C2D-based F2F designs. The total area savings of F2F designs over the 2D is 57.8% for LDPC, and 53.0% for both AES-128, and JPEG. As shown in Table 8, our C2D flow offers a 20-34% wirelength savings and a 4-13% standard cell area savings. Therefore, the wire-dominated LDPC, which shows the highest wirelength to standard cell area ratio, benefits most from C2D in terms of the total power savings at iso-performance (26.8%), whereas the lowest wirelength to standard cell area ratio benchmark, JPEG, gains the lowest total power savings (5.7%). An interesting trend is that the standard cell area reduction depends on the ratio of sequential cell count to the total number of cells. Since the number of sequential cells in a design is not changed, only reduced drive strength for the sequential cells contributes to the power savings. On the other hand, buffers are optimized in both number and strength. Therefore, LDPC, which has the lowest sequential cell count to the total cell count ratio (2.7%), achieves the largest standard cell area savings (12.7%) in the F2F design.

Table 8: Iso-performance power comparison between commercial 2D vs. C2D. Δ% indicates the savings over 2D designs.

Design	2D	C2D	Δ%
LDPC, 2GHz			
Footprint ($\mu m \times \mu m$)	555.7 × 555.1	361.2 × 360.8	57.8
F2F Via Count	-	21,575	-
Cell Count	77,024	64,610	16.1
Seq. Cell Count (%)	2,048 (2.7%)	2,048 (3.2%)	0.0
Standard Cell Area (μm^2)	204,782	178,876	**12.7**
Total Wirelength (m)	3.8	2.5	**33.6**
Tot. WL / Cell Area (m^{-1})	**18.7**	14.2	24.1
Switching Power (mW)	193.9	136.9	29.4
Cell Internal Power (mW)	33.0	28.8	12.7
Leakage Power (mW)	11.1	8.2	26.1
Total Power (mW)	237.8	174.0	**26.8**
AES-128, 5.4GHz			
Footprint ($\mu m \times \mu m$)	716 × 715.6	490.9 × 490.6	53.0
F2F Via Count	-	63,211	-
Cell Count	147,483	140,960	4.4
Seq. Cell Count (%)	10,688 (7.2%)	10,688 (7.6%)	0.0
Standard Cell Area (μm^2)	377,702	361,096	**4.4**
Total Wirelength (m)	2.9	2.2	**22.9**
Tot. WL / Cell Area (m^{-1})	**7.7**	6.2	19.5
Switching Power (mW)	250.8	223.7	10.8
Cell Internal Power (mW)	113.6	108.4	4.6
Leakage Power (mW)	17.5	16.1	8.0
Total Power (mW)	381.9	348.2	**8.8**
JPEG, 2.16GHz			
Footprint ($\mu m \times \mu m$)	1156.3 × 1153.7	792.8 × 791.0	53.0
F2F Via Count	-	121,357	-
Cell count	312,451	284,884	8.8
Seq. Cell Count (%)	37,538 (12.0%)	37,538 (13.2%)	0.0
Standard Cell Area (μm^2)	982,231	943,812	**3.9**
Total Wirelength (m)	5.8	4.6	**20.2**
Tot. WL / Cell Area (m^{-1})	**5.9**	4.9	16.9
Switching Power (mW)	415.8	385.9	7.2
Cell Internal Power (mW)	195.1	189.9	2.7
Leakage Power (mW)	30.2	28.5	5.6
Total Power (mW)	641.1	604.4	**5.7**

6 CONCLUSIONS

To maximize the utilization of 3D interconnect and the power-performance-area benefit of F2F-bonded 3D ICs, in this paper, we proposed a full-chip RTL-to-GDSII physical design solution named Compact-2D (C2D) that offers a commercial-quality F2F-bonded 3D IC physical layout. We presented interconnect RC scaling, placement contraction, and memory expansion idea, which allows us to utilize the original technology files and design rules of the target technology node for a F2F-bonded 3D IC implementation. We also introduced placement row splitting idea to enable post-tier-partitioning optimization in our C2D flow, which is completely missing in the state-of-the-art F2F physical design solution. With our extensive experiments and analysis, we evaluated the impact of those ideas in the final F2F design results, and showed that using 28nm process design kit, F2F-bonded 3D ICs implemented by our C2D flow offers a maximum 26.8% of total power reduction with a maximum 15.6% silicon area savings compared to the 2D IC designs at iso-performance.

For the future directions, we will generalize C2D flow to handle more than two tiers. For example, we can adjust the scaling factors in interconnect RC scaling / placement contraction for multi-tier designs. Various multi-way balanced partitioning schemes can be applied to the tier partitioning, and placement row splitting in the Compact F2F via planning can be based on the number of dies. Lastly, we can adopt C2D to build commercial-quality TSV-based 3D ICs or monolithic 3D ICs as well with advanced tier partitioning algorithms for the given implementation and fabrication constraints. All these challenges are the future works for C2D flow.

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